

FIE401: Financial Econometrics

Panel Data Analysis

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Basics of Panel Data

Panel Data (i)

- ▶ A panel dataset contains observations on multiple entities (individuals, states, companies, . . .), where each entity is observed at two or more points in time.
- ▶ Hypothetical examples:
 - ▶ Data on 50 U.S. states, each state is observed in 3 years, for a total of 150 observations.
 - ▶ Data on 1000 firms, in ten different years, for 10,000 observations total.

Panel Data (ii)

- ▶ Notation:
 - ▶ double subscript distinguishes entities and time periods.
 - ▶ i = entity (e.g., firm), N = number of entities, so $i = 1, \dots, N$.
 - ▶ t = time period (e.g., year), T = number of time periods, so $t = 1, \dots, T$.
- ▶ Panel data with K regressors: $(x_{1it}, x_{2it}, \dots, x_{Kit}, y_{it})$,
 $i = 1, \dots, N$, $t = 1, \dots, T$.
- ▶ Panel data = Longitudinal data
- ▶ Balanced data = no missing observations, i.e., all variables are observed for all entities (states) and all time periods (years).
- ▶ In Finance, we usually deal with unbalanced panels.

Why are panel data useful?

- ▶ With panel data we can control for factors that:
 - ▶ Vary across entities but do not vary over time;
 - ▶ Could cause omitted variable bias if they are omitted;
 - ▶ Are unobserved or unmeasured – and therefore cannot be included in the regression using multiple regression.
- ▶ The key idea: If an omitted variable does not change over time, then any changes in y over time cannot be caused by the omitted variable.

Example: Drunk Driving Laws and Traffic Deaths

- ▶ Approx. 40,000 traffic fatalities annually in the U.S.
- ▶ 1/3 of traffic fatalities involve a drinking driver.
- ▶ 25% of drivers on the road between 1 a.m. and 3 a.m. have been drinking (estimate).
- ▶ A drunk driver is 13 times as likely to cause a fatal crash as a non-drinking driver (estimate).
- ▶ Public policy issues:
 - ▶ Drunk driving causes massive externalities (sober drivers are killed, society bears medical costs, etc.).
 - ▶ Are there any effective ways to reduce drunk driving?
 - ▶ What are effects of specific laws, e.g., mandatory punishment, minimum legal drinking age, economic interventions (alcohol taxes)?

Example: Drunk Driving Laws and Traffic Deaths

- ▶ Observational unit: a year in a U.S. state.
 - ▶ 48 U.S. states, so $N = \# \text{ of entities} = 48$.
 - ▶ 7 years (1982, . . . , 1988), so $T = \# \text{ of time periods} = 7$.
 - ▶ Balanced panel, so total $\# \text{ observations} = 7 * 48 = 336$
- ▶ Variables:
 - ▶ Traffic fatality rate, $mrall$ ($\#$ traffic deaths in that state in that year, per 10,000 state residents).
 - ▶ Tax on a case of beer, $beertax$.
 - ▶ Other: legal driving age ($mlda$), mandatory jail sentence ($jaild$), average miles per driver ($vmiles$), and mandatory community service ($comserd$).

Pooled OLS

Example: Drunk Driving Laws and Traffic Deaths

$$mrall_{it} = \beta_0 + \beta_1 beertax_{it} + u_{it}$$

$$\widehat{mrall}_{it} = 1.85 + 0.36 \times beertax_{it}$$

```
require(plm);  
#load data  
load("M:/Projects/FIE401/data/data_lectures/Traffic.Rdata");  
#define panel data  
Traffic<-pdata.frame(Traffic,index=c("state","year"));  
#pooled OLS using lm  
fit1<-lm(mrall~beertax,data=Traffic);  
summary(fit1);  
#pooled OLS using plm  
fit2<-plm(mrall~beertax,data=Traffic,model="pooling");  
summary(fit2);
```

Example: Drunk Driving Laws and Traffic Deaths

```
fit1<-lm(mrall~beertax,data=Traffic);
```

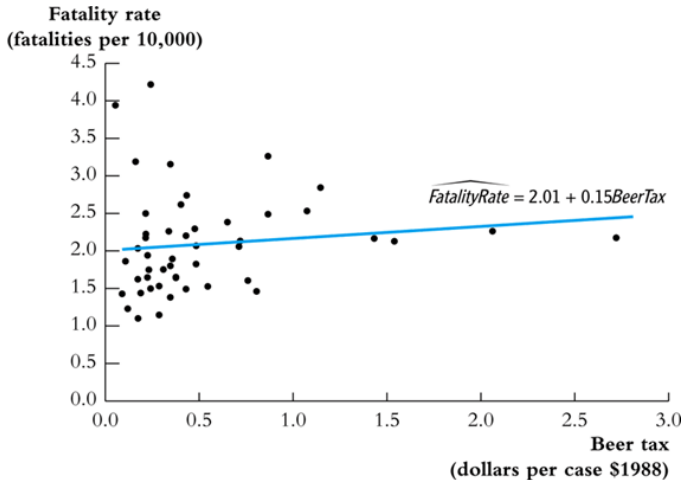
```
##
## Call:
## lm(formula = mrall ~ beertax, data = Traffic)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.09060 -0.37768 -0.09436  0.28548  2.27643
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.85331     0.04357  42.539 < 2e-16 ***
## beertax      0.36461     0.06217   5.865 1.08e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5437 on 334 degrees of freedom
## Multiple R-squared:  0.09336,    Adjusted R-squared:  0.09065
## F-statistic: 34.39 on 1 and 334 DF,  p-value: 1.082e-08
```

Example: Drunk Driving Laws and Traffic Deaths

```
fit2<-plm(mrall~beertax,data=Traffic,model="pooling");

## Pooling Model
##
## Call:
## plm(formula = mrall ~ beertax, data = Traffic, model = "pooling")
##
## Balanced Panel: n = 48, T = 7, N = 336
##
## Residuals:
##      Min.   1st Qu.   Median   3rd Qu.    Max.
## -1.09060 -0.37768 -0.09436  0.28548  2.27643
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## (Intercept)  1.853308   0.043567  42.5391 < 2.2e-16 ***
## beertax      0.364605   0.062170   5.8647 1.082e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    108.92
## Residual Sum of Squares: 98.747
```

Example: Drunk Driving Laws and Traffic Deaths



(a) 1982 data

Example: Drunk Driving Laws and Traffic Deaths

- ▶ Why might there be higher more traffic deaths in states that have higher alcohol taxes?
- ▶ Suppose:
 - ▶ High traffic density means more traffic deaths.
 - ▶ States with lower traffic density have lower alcohol taxes.
 - ▶ Then the two conditions for omitted variable bias are satisfied.
 - ▶ Specifically, “high taxes” could reflect “high traffic density” (so the OLS coefficient would be biased positively - high taxes, more deaths).
 - ▶ Panel data lets us eliminate omitted variable bias when the omitted variables are constant over time within a given state.

First Difference Estimator

Panel Data with Two Time Periods (i)

$$mrall_{it} = \beta_0 + \beta_1 beertax_{it} + \beta_2 z_i + u_{it}$$

- ▶ z_i is a factor that does not change over time (density), at least during the years on which we have data.
- ▶ Suppose z_i is not observed, and hence cannot be controlled for in multiple regression model.
- ▶ Omission of z_i could result in omitted variable bias.
- ▶ The effect of z_i can be eliminated using $T = 2$ years.

Panel Data with Two Time Periods (ii)

$$mrall_{it} = \beta_0 + \beta_1 beertax_{it} + \beta_2 z_i + u_{it}$$

- ▶ Any change in the fatality rate ($mrall$) from 1982 to 1988 cannot be caused by z_i , because z_i (by assumption) does not change between 1982 and 1988.

$$mrall_{i,1988} = \beta_0 + \beta_1 beertax_{i,1988} + \beta_2 z_i + u_{i,1988}$$

$$mrall_{i,1982} = \beta_0 + \beta_1 beertax_{i,1982} + \beta_2 z_i + u_{i,1982}$$

- ▶ Suppose $E(u_{it} | beertax_{it}, z_i) = 0$.
- ▶ Subtracting 1988 - 1982 (that is, calculating the change), eliminates the effect of z_i .

Panel Data with Two Time Periods (iii)

$$mrall_{i,1988} = \beta_0 + \beta_1 beertax_{i,1988} + \beta_2 z_i + u_{i,1988}$$

$$mrall_{i,1982} = \beta_0 + \beta_1 beertax_{i,1982} + \beta_2 z_i + u_{i,1982}$$

$$\begin{aligned} mrall_{i,1988} - mrall_{i,1982} = \\ \beta_1 (beertax_{i,1988} - beertax_{i,1982}) + (u_{i,1988} - u_{i,1982}) \end{aligned}$$

- ▶ The new error term, $(u_{i,1988} - u_{i,1982})$, is uncorrelated with either $beertax_{i,1982}$ or $beertax_{i,1988}$.
- ▶ This “difference” equation can be estimated by OLS, even though z_i is not observed.
- ▶ The omitted variable z_i does not change, so it cannot be a determinant of the change in $mrall$.
- ▶ This differences regression does not have an intercept – it was eliminated by the subtraction step.

Example: Drunk Driving Laws and Traffic Deaths

$$\widehat{mrall}_{i,1982} = 2.01 + 0.15 \times beertax_{i,1982}$$

$$\widehat{mrall}_{i,1988} = 1.86 + 0.44 \times beertax_{i,1988}$$

$$\widehat{mrall}_{i,1988-1982} = -0.87 \times beertax_{i,1988-1982}$$

#OLS for 1982 and 1988

```
fit1<-plm(mrall~beertax,  
          data=Traffic[(Traffic$year==1982),],  
          model="pooling");
```

```
fit2<-plm(mrall~beertax,  
          data=Traffic[(Traffic$year==1988),],  
          model="pooling");
```

#first difference method

```
fit3<-plm(mrall~0+beertax,  
          data=Traffic[(Traffic$year==1988  
                        | Traffic$year==1982),],  
          model="fd");
```

```
summary(fit3);
```

Example: Drunk Driving Laws and Traffic Deaths

```
## Oneway (individual) effect First-Difference Model
##
## Call:
## plm(formula = mrall ~ 0 + beertax, data = Traffic[(Traffic$year ==
##      1988 | Traffic$year == 1982), ], model = "fd")
##
## Balanced Panel: n = 48, T = 2, N = 96
## Observations used in estimation: 48
##
## Residuals:
##      Min. 1st Qu.  Median      Mean 3rd Qu.     Max.
## -1.3004 -0.1619   0.0266 -0.0634   0.1595   0.6136
##
## Coefficients:
##              Estimate Std. Error t-value Pr(>|t|)
## beertax -0.86892      0.39299 -2.2111  0.03193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:      8.1082
## Residual Sum of Squares: 7.3608
## R-Squared:      0.11919
```

Fixed effects

Entity Fixed Effects Regression (i)

What if you have more than 2 time periods ($T > 2$)?

$$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 z_i + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

We can rewrite this in two equivalent ways:

- ▶ “Binary regressor” model.
- ▶ “Fixed Effects” model.

Entity Fixed Effects Regression (ii)

Suppose we have $N = 3$ states: California, Texas, and Massachusetts.

$$mrall_{it} = \beta_0 + \beta_1 beertax_{it} + \beta_2 z_i + u_{it}, \quad i = 1, \dots, 3, \quad t = 1, \dots, T$$

Population regression for California state:

$$mrall_{CA,t} = \beta_0 + \beta_1 beertax_{CA,t} + \beta_2 z_{CA} + u_{CA,t}$$

$$mrall_{CA,t} = (\beta_0 + \beta_2 z_{CA}) + \beta_1 beertax_{CA,t} + u_{CA,t}$$

$$mrall_{CA,t} = \alpha_{CA} + \beta_1 beertax_{CA,t} + u_{CA,t}$$

- ▶ $\alpha_{CA} = \beta_0 + \beta_2 z_{CA}$ does not change over time.
- ▶ α_{CA} is the intercept for California state and β_1 is the regression slope.

Entity Fixed Effects Regression (iii)

Suppose we have $n = 3$ states: California, Texas, and Massachusetts.

$$mrall_{CA,t} = \alpha_{CA} + \beta_1 beertax_{CA,t} + u_{CA,t}$$

$$mrall_{TX,t} = \alpha_{TX} + \beta_1 beertax_{TX,t} + u_{TX,t}$$

$$mrall_{MA,t} = \alpha_{MA} + \beta_1 beertax_{MA,t} + u_{MA,t}$$

OR

$$mrall_{i,t} = \alpha_i + \beta_1 beertax_{i,t} + u_{i,t}$$

- ▶ The intercept is unique for each state, but the slope is the same in all the states: parallel lines.
- ▶ Recall that shifts in the intercept can be represented using binary regressors.

Entity Fixed Effects Regression (iv)

Binary regressors form:

$$mrall_{i,t} = \beta_0 + \gamma_{CA}DCA_i + \gamma_{TX}DTX_i + \beta_1 beertax_{i,t} + u_{i,t}$$

- ▶ $DCA_i = 1$ if state is California, and 0 otherwise.
- ▶ $DTX_i = 1$ if state is Texas, and 0 otherwise.
- ▶ Why do we leave out DMA_i ?

Entity Fixed Effects Regression (v)

- ▶ Binary regressors form:

$$y_{i,t} = \beta_0 + \beta_1 x_{i,t} + \gamma_2 D2_i + \gamma_3 D3_i + \dots + \gamma_N DN_i + u_{i,t}$$

where $D2_i = 1$ if $i = 2$ and 0 otherwise

- ▶ Entity fixed effects form:

$$y_{i,t} = \alpha_i + \beta_1 x_{i,t} + u_{i,t}$$

where α_i is called "entity fixed effect"

Estimation of Entity Fixed Effects Regression

- ▶ Three equivalent estimation methods:
 - ▶ Binary OLS regression.
 - ▶ “Entity-demeaned” OLS regression.
 - ▶ “First difference” specification without intercept (only works for $T = 2$).
- ▶ We already did the “first difference” specification (1988 minus 1982) – but this only works for $T = 2$ years
- ▶ Methods #1 and #2 work for general T .
- ▶ Method #1 is only practical when N is not too large.

Example: Drunk Driving Laws and Traffic Deaths

Method # 1: Use factor operator

$$\widehat{mrall}_{i,t} = 3.48 - 0.66 \times beertax_{i,t} + \gamma_2 D2_i + \gamma_3 D2_i + \dots + \gamma_N DN_i$$

```
#load data
load("M:/Projects/FIE401/data/data_lectures/Traffic.Rdata");
#Binary OLS regression
fit<-lm(mrall~beertax+factor(state),data=Traffic);
summary(fit);
```

Example: Drunk Driving Laws and Traffic Deaths

Method # 1: Use factor operator

```
##  
## Call:  
## lm(formula = mra11 ~ beertax + factor(state), data = Traffic)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.58696 -0.08284 -0.00127  0.07955  0.89780   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    3.47763    0.31336  11.098 < 2e-16 ***  
## beertax        -0.65587    0.18785  -3.491 0.000556 ***  
## factor(state)AR -0.65495    0.21902  -2.990 0.003028 **  
## factor(state)AZ -0.56773    0.26667  -2.129 0.034107 *  
## factor(state)CA -1.50947    0.30435  -4.960 1.21e-06 ***  
## factor(state)CO -1.48428    0.28735  -5.165 4.50e-07 ***  
## factor(state)CT -1.86226    0.28053  -6.638 1.58e-10 ***  
## factor(state)DE -1.30760    0.29395  -4.448 1.24e-05 ***  
## factor(state)FL -0.26813    0.13933  -1.924 0.055284 .  
## factor(state)GA  0.52460    0.18395   2.852 0.004661 **  
## factor(state)IA -1.54393    0.25344  -6.092 3.58e-09 ***
```

Example: Drunk Driving Laws and Traffic Deaths

Method # 2: Use plm package

$$\widehat{mrall}_{i,t} = \alpha_i - 0.66 \times beertax_{i,t}$$

```
require(plm);  
#load data  
load("M:/Projects/FIE401/data/data_lectures/Traffic.Rdata");  
#define panel data  
Traffic<-pdata.frame(Traffic,index=c("state","year"));  
# Entity-demeaned regression  
fit<-plm(mrall~beertax,data=Traffic,  
         effect="individual",model="within");  
summary(fit);  
#extract fixed effects  
fixef(fit,effect="individual");
```

Example: Drunk Driving Laws and Traffic Deaths

Method # 2: Use plm package

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = mrall ~ beertax, data = Traffic, effect = "individual"
##      model = "within")
##
## Balanced Panel: n = 48, T = 7, N = 336
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.5869619 -0.0828376 -0.0012702  0.0795455  0.8977960
##
## Coefficients:
##      Estimate Std. Error t-value Pr(>|t|)
## beertax -0.65587    0.18785 -3.4915 0.000556 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    10.785
## Residual Sum of Squares: 10.345
## R-Squared:    0.040745
```

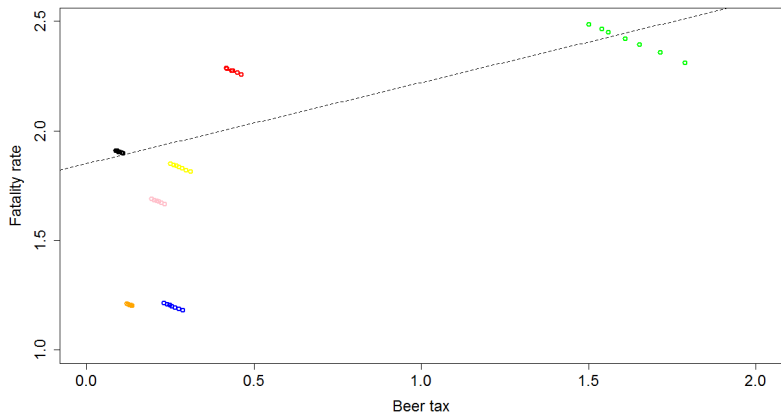
Example: Drunk Driving Laws and Traffic Deaths

Method # 2: Use plm package

```
#extract fixed effects  
fixef(fit,effect="individual");
```

##	AL	AR	AZ	CA	CO	CT	DE	
##	3.477630	2.822678	2.909903	1.968161	1.993350	1.615373	2.170028	3.209
##	GA	IA	ID	IL	IN	KS	KY	
##	4.002233	1.933698	2.808608	1.516008	2.016088	2.254414	2.260113	2.630
##	MA	MD	ME	MI	MN	MO	MS	
##	1.367884	1.771190	2.369683	1.993103	1.580417	2.181368	3.448550	3.117
##	NC	ND	NE	NH	NJ	NM	NV	
##	3.187165	1.854191	1.955452	2.223176	1.371881	3.904005	2.876855	1.290
##	OH	OK	OR	PA	RI	SC	SD	
##	1.803211	2.932569	2.309630	1.710164	1.212576	4.034804	2.473909	2.601
##	TX	UT	VA	VT	WA	WI	WV	
##	2.560157	2.313680	2.187447	2.511586	1.818106	1.718364	2.580876	3.249

Example: Drunk Driving Laws and Traffic Deaths



Time Fixed Effects Regression

- ▶ An omitted variable might vary over time but not across states.
- ▶ Safer cars (air bags, etc.), changes in national laws, ...
- ▶ These produce intercepts that change over time.
- ▶ Let s_t denote the combined effect of variables which changes over time but not states (“safer cars”).
- ▶ The resulting population regression model is:

$$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_3 s_t + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

Time Fixed Effects Regression

Binary regressors form:

$$y_{i,t} = \beta_0 + \beta_1 x_{i,t} + \delta_2 B2_t + \delta_3 B3_t + \dots + \delta_T BT_t + u_{i,t}$$

where $B2_t = 1$ if $t = 2$ and 0 otherwise

Time fixed effects form:

$$y_{i,t} = \lambda_t + \beta_1 x_{i,t} + u_{i,t}$$

where λ_t is called "time fixed effect"

Two equivalent estimation methods:

- ▶ Binary OLS regression.
- ▶ "Time-demeaned" OLS regression.

Two-way Fixed Effects regression

$$y_{i,t} = \alpha_i + \lambda_t + \beta_1 x_{i,t} + u_{i,t}$$

- ▶ When $T = 2$, computing the first difference and including an intercept is equivalent to (gives exactly the same regression as) including entity and time fixed effects.
- ▶ When $T > 2$, there are various equivalent ways to incorporate both entity and time fixed effects:
 - ▶ Entity-demeaning and $T - 1$ time indicators.
 - ▶ Time-demeaning and $N - 1$ entity indicators.
 - ▶ $T - 1$ time indicators and $N - 1$ entity indicators.
 - ▶ Entity- and time-demeaning.

Example: Drunk Driving Laws and Traffic Deaths

$$\widehat{mrall}_{i,t} = \alpha_i + \lambda_t - 0.64 \times beertax_{i,t}$$

Method: Entity- and time-demeaning.

```
require(plm);  
#load data  
load("M:/Projects/FIE401/data/data_lectures/Traffic.Rdata");  
#define panel data  
Traffic<-pdata.frame(Traffic,index=c("state","year"));  
# Entity- and time-demeaned regression  
fit4<-plm(mrall~beertax,data=Traffic,  
          effect="twoway",model="within");  
summary(fit4);  
fixef(fit4,effect="individual");  
fixef(fit4,effect="time");
```

Example: Drunk Driving Laws and Traffic Deaths

Method: Entity- and time-demeaning.

```
## Twoways effects Within Model
##
## Call:
## plm(formula = mrall ~ beertax, data = Traffic, effect = "twoway",
##      model = "within")
##
## Balanced Panel: n = 48, T = 7, N = 336
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.5955623 -0.0809618  0.0014301  0.0823355  0.8388350
##
## Coefficients:
##           Estimate Std. Error t-value Pr(>|t|)
## beertax -0.63998    0.19738  -3.2424 0.001328 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    10.29
## Residual Sum of Squares: 9.9193
## R-Squared:    0.036065
```

Assumptions and Inference

Least Squares Assumptions

- ▶ LS1. $E(u_{it} | x_{i1}, \dots, x_{iT}, \alpha_i) = 0$
- ▶ LS2. $(x_{i1}, \dots, x_{iT}, u_{i1}, \dots, u_{iT})$, for $i = 1, \dots, N$ are i.i.d.
- ▶ LS3. (x_{it}, u_{it}) have finite fourth moments (no outliers).
- ▶ LS4. There is no perfect multicollinearity.

$$\text{LS1. } E(u_{it} | x_{i1}, \dots, x_{iT}, \alpha_i) = 0$$

- ▶ u_{it} has mean zero, given the entity fixed effect and the entire history of the x 's for that entity.
- ▶ This means there are no omitted lagged effects (any lagged effects of x must enter explicitly).
- ▶ Also, there is not feedback from u to future x . I.e., whether a state has a particularly high fatality rate this year doesn't subsequently affect whether it increases the beer tax.

LS2. $(x_{i1}, \dots, x_{iT}, u_{i1}, \dots, u_{iT})$ are i.i.d.

- ▶ It is satisfied if entities are randomly sampled from their population by simple random sampling.
- ▶ This does not require observations to be i.i.d. over time for the same entity – that would be unrealistic.
- ▶ Whether a state has a high beer tax this year is a good predictor of (correlated with) whether it will have a high beer tax next year.
- ▶ Similarly, the error term for an entity in one year is plausibly correlated with its value in the year, that is, $\text{corr}(u_{it}, u_{i,t+1})$ is often plausibly nonzero.

Inference

- ▶ The OLS fixed effect estimator is unbiased, consistent, and asymptotically normally distributed.
- ▶ However, the usual OLS standard errors will in general be wrong because they assume that u_{it} is serially uncorrelated.
- ▶ Solution: clustered standard errors.
 - ▶ Allow for autocorrelation **within** entities.
 - ▶ Are also robust for heteroskedasticity within and across entities.

Example: Drunk Driving Laws and Traffic Deaths

$$\widehat{mrall}_{i,t} = \alpha_i - 0.66 \times beertax_{i,t}$$

```
require(plm);
require(lmtest);
require(sandwich);
#load data
load("M:/Projects/FIE401/data/data_lectures/Traffic.Rdata");
#define panel data
Traffic<-pdata.frame(Traffic,index=c("state","year"));
# Entity-demeaned regression
fit<-plm(mrall~beertax,data=Traffic,
         effect="individual",model="within");
#SEs clustered by entity
coeftest(fit,vcov=vcovHC(fit,cluster="group",type="sss"));
```

Example: Drunk Driving Laws and Traffic Deaths

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## beertax -0.65587      0.29142 -2.2506  0.02517 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Summary: Panel Data Regressions (i)

$$y_{i,t} = \alpha_i + \lambda_t + \beta_1 x_{i,t} + u_{i,t}$$

- ▶ Advantages:
 - ▶ Control for unobserved variables that vary across entities but not over time.
 - ▶ Control for unobserved variables that vary over time but not across entities.
 - ▶ More observations give you more information.
 - ▶ Estimation involves relatively straightforward extensions of multiple regression.
 - ▶ Statistical inference like in multiple regression.

Summary: Panel Data Regressions (ii)

$$y_{i,t} = \alpha_i + \lambda_t + \beta_1 x_{i,t} + u_{i,t}$$

- ▶ Limitations:
 - ▶ Need variation in x over time within entities.
 - ▶ Time lag effects can be important – we did not model those in the beer tax application but they could matter (dynamic panels: Arellano-Bond estimator).
 - ▶ You need to use clustered standard errors to guard against the often-plausible possibility of autocorrelation in u_{it} .

Textbook

- ▶ Panel data analysis: Chapter 10, Stock and Watson, Introduction to Econometrics, Global Edition, 4th edition